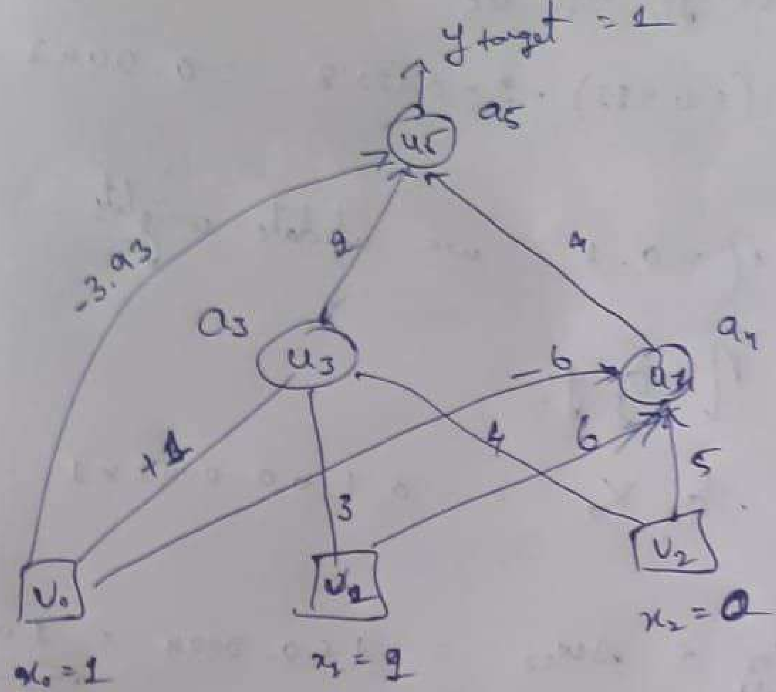
29/10/25

ANN DL




At $a_3$,

$a_3 = 1 \cdot 1 + 1 \times 3 + 0 \times 4 = \underline{4} \Rightarrow f(a_3) = \frac{1}{1+e^{-4}} = 0.982$

At $a_4$, $a_4 = -6 \times 1 + 1 \times 6 + 0 \times 5 = \underline{0} \Rightarrow f(a_4) = \frac{1}{1+e^{0}} = 0.5$

At $a_5$,

$a_5 = -3.93 \times 1 + 0.982 \times 2 + 0.5 \times 4 = \underline{0.034}$

$\Rightarrow f(a_5) = \frac{1}{1+e^{-0.034}} = \underline{0.5084}$

Error, $1 - 0.5084 = \underline{0.4916}$

target

$\delta_k = (t_k - a_k)\, a_k\, (1 - a_k)$

$\delta_5 = (1 - 0.5084)\, 0.5084\, (1 - 0.5084) = \underline{0.1228}$

$\delta_4 = y_4\, (1 - y_4)\, \omega_{45}\, \delta_5$

$= 0.5\, (1 - 0.5)\, 4 \times 0.1228 = 0.1228$

$$\delta_3 = y_3\,(1-y_3)\,w_{35}\,\delta_5$$
$$= 0.982\,(1-0.982)\cdot 2 \times 0.1228 = 0.0043$$

Learing rate $\eta = 0.1$, we update weights

$$\Delta w_{ij} = \eta\,\delta_j\,y_i$$

$$\Delta w_{03} = \eta\cdot\delta_3\,x_0 = 0.1 \times 0.0043 \times 1 = 0.00043$$

$$w_{03_{new}} = w_{03_{old}} + \Delta w_{03} = 1 + 0.0004 = \underline{\underline{1.0004}}$$

$$w_{04} = w_{04_{old}} + \eta\,\delta_4\,x_0 = -6 + 0.1 \times 0.1228 \times 1$$
$$= \underline{\underline{-5.9877}}$$

$$w_{05} = w_{05_{old}} + \eta\,\delta_5\,x_0 = -3.93 + 0.1 \times 0.1228 \times 1$$
$$= -3.9177$$

$$w_{13} = w_{13_{old}} + \eta\,\delta_3\,x_1 = 3 + 0.1 \times 0.0043 \times 1$$
$$= \underline{\underline{3.00043}}$$

$$w_{14} = w_{14_{old}} + \eta\,\delta_4\,x_1 = 6 + 0.1 \cdot 0.1228 \cdot 1$$
$$= \underline{\underline{6.01228}}$$

$$w_{23} = w_{23_{old}} + \eta\,\delta_3\,x_3 = 4 + 0.1 \cdot 0.0043 \cdot 0 = \underline{\underline{4}}$$

$$w_{24} = \text{———} = 5 + 0 = \underline{\underline{5}}$$

$$w_{35} = w_{35_{old}} + \eta\,\delta_5\,y_3 = 2 + 0.1 \times 0.1228 \times 0.982 = \underline{\underline{2.0120}}$$

$$w_{45} = w_{45_{old}} + \eta\,\delta_5\,y_4 = 4 + (0.1 \times 0.1228 \times 0.5) = \underline{\underline{4.00614}}$$

Forward pass

At $a_3$,

$a_3 = 1.0004 + 3.00043 + 0 = 4.00083 \Rightarrow f(a_3) = y_3 = 0.9820$

At $a_4$

$a_4 = 1 \cdot (-5.9177) + 1 \cdot 6.01228 = \quad \Rightarrow y_4 = 0.5061$

$\Rightarrow$ At $a_5$

$a_5 = 1 \cdot (-3.9177) + 0.9820 \times 2.0120 + 0.5061 \cdot 4.00614$

$\Rightarrow y_5 = \underline{\underline{0.5213}}$

$\therefore$ Error $= 1 - 0.52137$

$= \underline{\underline{0.4786}}$